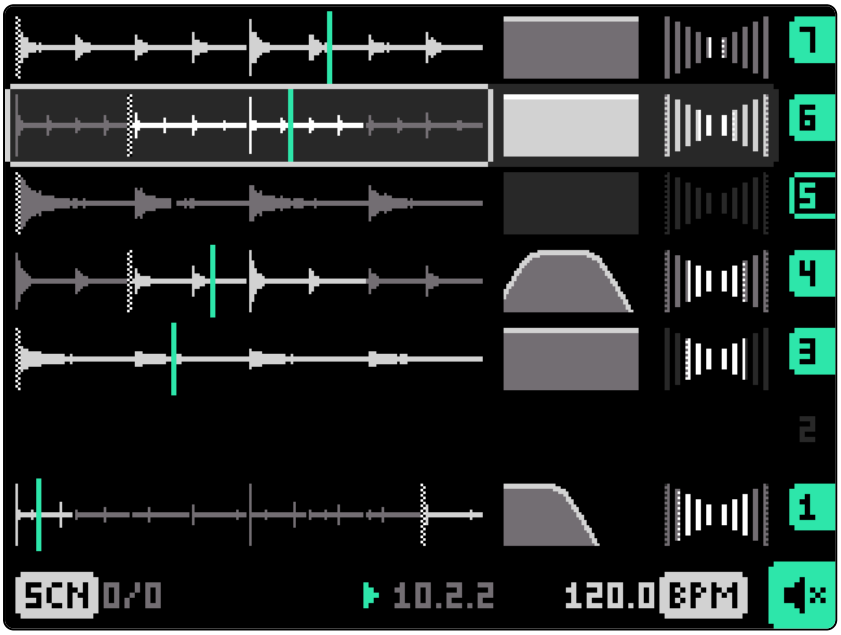
L7

A 7-track looper for the Dirtywave M8. Quick user's guide.

L7 replaces the M8's own firmware with a live looper. You record loops from the M8's line input onto 7 tracks, shape each one (range, filter, volume, balance), and store combinations as **scenes**. You keep up to 12 **projects** on the SD card, and you can export any of them as an Ableton Live set.

The M8's keys are named on the device: the arrows, `SHIFT`, `START`, `B` (OPT), `A` (EDIT) and the power key. This guide uses those names.

The screen



Track rows: tracks 7 (top) down to 1, just above the status row. Each row has three cells and a button.

Waveform: the recorded loop; the bright part is the range that plays, the line is the playhead.

Filter: the response of the track's highpass and lowpass filter. As you narrow it, L7 turns the track up to make up for about half of what the filter takes away (at most 6 dB), worked out from each recording's own mix of lows and highs: a bass line under a lowpass isn't turned up, a hi-hat loop is. A limiter on the master keeps the mix (and what you resample) from clipping.

Volume: the track's level and balance, with a level meter. Waveforms are drawn as if normalised (every clip's loudest point near the full height), and get smaller as you turn the track down.

Button (right): the track number, filled while the track plays, outlined while it's muted, dim while it's empty.

Status row (bottom): the scene (`SCN 3/7`), the clock (bar.beat.step), the tempo, and the **action button**, which always shows what `A` will do.

The selected cell has a light frame. Move it with the arrows. The colour accents follow the loaded project's colour.

Keys at a glance

KEYS	WHAT THEY DO
Arrows	Move the selection: `UP`/`DOWN` between rows, `LEFT`/`RIGHT` between cells.

A on a track	Empty track: record, press again to stop. Track with a clip: mute / unmute. Keep **A** held and press (or hold) **UP** / **DOWN** to **swipe**: each track you step onto gets the same, and it stops when it comes back round to where you started. Step the other way to undo, a track at a time.
B held + arrows	Adjust the selected cell (see below).
B twice	Reset the cell: whole loop, filter open, volume at unity and balance centred; on the clock cell, master volume back to 100% and balance centred; on the tempo cell, the scene's own tempo (the one it was made at, or its latest recording's).
SHIFT + **A**	Cut the track's clip to the clipboard, or paste it onto an empty track. On a recording track: cancel the recording.
SHIFT + **LEFT** / **RIGHT**	On a track: put the previous / next recording on the card on the track (in name order; while you hold **SHIFT** the track's waveform is outlined in the accent colour, with the recording's name and number). On an empty track, **LEFT** brings back the project's last recording you used, **RIGHT** the one after it. Holding **SHIFT** and **B** together, the arrows do what the one you pressed last does. On the status row: nudge the clock faster / slower while you hold it, like Live's nudge buttons: the tempo comes back when you let go, the shift stays (following an external clock, the nudge moves L7's offset from it). Following one, **B** + **LEFT** / **RIGHT** on the tempo cell moves L7 10 ms earlier / later against it (±250 ms, kept for each source; both arrows: back to 0), and **B** + **UP** / **DOWN** doubles / halves the tempo taken from it.
SHIFT + **UP** / **DOWN**	Next / previous scene.
START	Play from 1.1.1 / stop.
SHIFT + **START**	Stopped: carry on from where the clock last stopped (back to 1.1.1 once you load a project). Playing: restart from 1.1.1.
B + **A** on an empty track	Resample: record what you hear onto this track (as soon as you press **A**).
LEFT / **RIGHT** past a row's first / last cell, **UP** on track 7, or **DOWN** on the status row	Open the menu (projects, export, storage mode), once you let go.
LEFT + **RIGHT** pressed together and held	Switch between the looper and the menu, from anywhere.
Power key (hold)	Switch off. Everything is saved first.

Recording

1. Plug your source into the M8's line input. The input is always heard.
2. Select an empty track and press **A**; the recording is set going as you press it. If the clock is stopped, it starts and recording begins on bar 1. If it's running, recording starts on the next bar line. If you press within 0.1 seconds after a bar line, it starts on that bar line instead (counted from when you pressed, as long as you let go soon after).

3. Press **A** again to stop. Loops are always whole bars. If you stop within 0.1 seconds after a bar line, the loop ends there; otherwise recording carries on to the end of that bar. The longest loop is 32 bars.
4. The loop starts playing as soon as it's done, with no gap.

Pressing **A** on another empty track while one records stops the first and queues the next one to start exactly where the first ends. In a new, empty project a metronome click plays until the first loop is complete.

Shaping a track (B held + arrows)

CELL	LEFT / RIGHT	UP / DOWN
Waveform	Move the range's start (hold to keep moving; it carries on past the clip's ends).	Make the range longer / shorter (¼ bar up to the whole loop).
Filter	Move the band	Widen / narrow the band
Volume	Balance	Volume (off, then steps of 1.875 dB, two to a half line, the second half of a pair dotted; each clip starts at its level as recorded, and the top step makes it peak at full scale)
Any track		With **B** + **A** held (press **A** while holding **B**): move a track with a clip up / down. The track in its way takes its place (wraps round from track 1 to 7)
Any cell, **B** + **A**	Tap to nudge the loop later (right) / earlier (left) by a quarter beat (the range, start point and bar lines move over the waveform); hold to nudge it in small steps instead (1 ms at a time, then 2, 4, and 10 ms as you keep holding). It stops at 0; both arrows together: back to 0. The waveform and the tempo cell (in ms) show it; you hear it on the next quarter beat	Move the track (as from any cell)
Scene cell		Next / previous scene
Clock cell	Master volume (UP/DOWN) and balance (LEFT/RIGHT), shown like a track's volume cell (each project its own; up to 400%)	
Tempo cell	±0.1 BPM	±1 BPM (60–240 BPM)

Changing the tempo re-pitches the loops, like a turntable. The tempo belongs to the project. You can change it while recording too: the new loop plays at the average tempo of what was recorded, in step with the clock over the whole loop. Exported to Ableton Live, it keeps a warp marker at every bar, so Live plays each bar as it was played.

The status row with A

- **Scene cell**: go to the next scene when you let go of A (on the next beat); with the clock stopped, it starts the clock instead (and with fewer than two scenes it does nothing). After the last scene it goes round to scene 1. SHIFT + A duplicates the scene (the same as SHIFT + B twice). Press A again while still holding SHIFT to clear the copy's tracks, for a fresh empty scene.
- **Clock cell**: play / stop.
- **Tempo cell**: tap tempo. From the 3rd tap on, the tempo follows your taps, and keeps getting more accurate the longer you keep tapping (pause for over a second or press another key to start over; the slowest tempo you can tap is 60 BPM); with the clock stopped, the 5th tap starts it from 1.1.1, and a fresh estimate starts from that tap (following audio, tapping doesn't change the tempo: tap along from the bar's beat one, and after two taps the clock's bar eases onto yours; while it plays otherwise, tapping sets the tempo and also eases the clock onto your taps: the first tap of a run is taken as the bar's beat one, and the clock runs a little faster or slower until its beat one lines up with it). B + A (on any cell of the status row) turns the metronome on or off (with *Empty project click* on in the settings, a project with no clips yet clicks anyway), and the button shows a metronome, crossed out while it would click (with the clock running). SHIFT + A turns **audio follow** on or off (below); holding SHIFT there, the button shows a jack plug (the input), crossed out while audio follow is on.

Scenes

A scene is a snapshot of all 7 tracks: which clip each plays, its range, mute, filter, volume and balance. Whatever you change updates the current scene as you go; there is no save step. A project holds up to 200 scenes.

- SHIFT + UP/DOWN steps through the scenes. While playing, the change happens on the next beat after you let go of SHIFT; the scene number flashes until then. Meanwhile the tracks show the scene you've picked, as if the clock were stopped and nothing selected (no playheads, meters or selection; once you let go and the change waits for its beat, the selection shows as a grey outline; the track numbers blink every half beat), and nothing can be changed: only the scene keys and START work. Going to 0* or N+1* also waits for you to let go of SHIFT, then happens on the next beat. Each loop carries on as far past its start point as the one before it on that track was, going round within its own length if it's shorter (a track that was empty starts from its start point).
- Below the first scene is **0***, an empty scene. Above the last is **N+1***, a copy of the last one. Changing either (its tempo too, once it holds clips) turns it into a real scene. Tap UP or DOWN three times quickly with SHIFT to jump to N+1* or 0*.
- SHIFT + B twice duplicates the scene.
- Cut a scene with SHIFT, then B, then A (in that order). Paste it after the current one with B, then SHIFT, then A. Cutting a scene with no clips removes it but leaves the clipboard as it was. Cutting 0* empties the clipboard, and pasting with an empty clipboard inserts an empty scene (the button shows the empty checkered square, as when clearing a duplicate). The cut takes you to the scene before, except for a scene you've just made by changing N+1*: cut before you leave it, it takes you back to N+1*.

Following a MIDI clock

Connect a MIDI clock source (a drum machine, a DAW through a MIDI interface) to the M8's MIDI in, or send clock to it over USB (it shows up as a MIDI device on your computer). While it plays (from its Start until its Stop), L7 follows it: the tempo cell shows its tempo, and you can't change the tempo by hand meanwhile. While it's stopped, the tempo is yours to change, even if it keeps sending clock. Its **Start** starts L7 from 1.1.1 (or restarts it), with the bars lined up with the source's; its **Stop** stops it. If it continues from somewhere in the song (Song Position, then Continue), or jumps while playing, L7 starts there too, on the source's next beat, the clock showing that bar and every loop playing from where it would be. L7 keeps its beat in step by running a touch faster or slower for a beat when it drifts. Following a clock over USB, it plays 2 ms ahead of it, so what it sends

back over USB audio lands on your DAW's beat. If you start L7 with START instead, the beats line up but not the bars. When the clock stops (or stops coming in), L7 keeps the average tempo it followed, ready for the next start, whether that's from the clock or from START. When the clock comes back at a clearly different tempo, L7 takes that one at once. L7 also sends its clock, Start and Stop, on the MIDI out and over USB, so other gear (or your DAW) can follow it. While a scene change waits for its beat (or a project change for its bar line), the metronome stays silent.

Audio follow

L7 can follow the tempo and beat of whatever plays into its input (the line input, or audio from your computer over USB). Select the tempo cell and press SHIFT + A to turn audio follow on (the button shows a jack plug (the input) while you hold SHIFT; crossed out while it's on, and SHIFT + A turns it off again). It's remembered when you switch off. After about 5 seconds of music L7 finds the tempo, then the beat: the tempo cell shows the tempo it hears and its number flashes on each beat: in black (blinking off) while it's still looking for the beat (before it has a tempo, at the tempo shown), and in the accent colour once it has locked onto it. When the music stops, L7 keeps the tempo it had (the number flashes black again); when it comes back, L7 finds the beat again after a few beats. Press START on a beat and L7 starts exactly on it; while it plays, it keeps its beat on the music's, following tempo changes too. Only the beats are matched: bar 1 is where you pressed START. While audio follow is on you can't set the tempo by hand, but if it picks half or double the tempo you want, hold B on the tempo cell and press UP to double it, DOWN to halve it (LEFT / RIGHT move L7 earlier / later against the music, 10 ms a press): the clock stays on the music's beat (halved, on every other one). You can do this before it has found the tempo too: it then looks for the tempo near the one shown. Turning audio follow on again starts from the tempo it hears. A MIDI clock being followed takes priority.

Following a MIDI clock or audio, the tempo it sets isn't saved in your scene: when you're back on L7's own clock (the MIDI clock stopped, or audio follow turned off), the scene's own tempo comes back. The doubling / halving (B + UP / DOWN) is remembered when you switch off, and goes back to the tempo as it comes in when the MIDI clock is unplugged (10 seconds without it) or audio follow is turned off.

USB audio

Connected to a computer, the M8 also shows up as an audio device called **L7**, with 16 channels to the computer and 2 from it. Channels 1 and 2 (Master) carry what you hear on the M8, master volume included, but without the metronome, so you can record L7 straight into your DAW. Channels 3 to 16 carry each track on its own, from the top row down: Ch 1 (the top row) on 3 and 4, Ch 2 on 5 and 6, and so on to Ch 7 (the bottom row) on 15 and 16. Each is the track as you hear it in the mix (its volume, balance, filter and mute) at full level whatever the master volume. Your DAW shows them by name ("Master L", "Ch 1 L" …), so you can record every track to its own channel at once. On Windows, if the PC has seen an older L7, remove it once in Device Manager (show hidden devices) so the new channels show up. Audio the computer plays to L7 is mixed with the line input: you can monitor it and record it into loops like anything on the line input. While audio comes in over USB, the line input is left out of the mix after a couple of seconds of silence (so an empty input adds no noise), and comes back the moment something plays into it.

Audio from the computer arrives a little late (its audio buffer, its sync settings), so L7 measures by how much: play your DAW's metronome (or anything with a clear hit on each beat) into L7 for a few bars while the clock runs, and from then on loops you record from USB land on the beat. Change the DAW's buffer size and let it measure again the same way.

Resampling

Select an empty track and press B + A. L7 records everything you hear (tracks and input, but not the master volume) from the next bar line (or, pressed within 0.1 seconds after one, from that one), like a normal recording. When it's saved, it moves to a new scene right after the current one, on track 1, and the clips that were muted

come along, still muted, on tracks 2, 3 and so on, in the same order. The action button shows a record dot with a plus while resampling is available.

Projects and the menu



Open the menu by pressing any arrow past the edge of the screen (**LEFT** or **RIGHT** past a row's ends, **UP** on track 7, **DOWN** on the status row; the a thick line along that edge of the screen shows while you hold the key). While the clock runs, the loaded project's card shows a small play icon. The clock keeps running. Go back with **RIGHT** or **LEFT** off the edge of the cards or on a list item, **UP** on the top row of cards, or **DOWN** on the last list item. Going back happens as soon as you press the arrow. Each screen keeps its own selection. Going off the top or bottom edge, you land on the other screen's bottom or top row, in the column you last used there; going off the left or right edge, you land on the other screen's far side, in the row you last used there. Keep holding the arrow and the switch happens after a moment (0.6 s) without letting go; the selection stays where it lands until you press again. With no project loaded, going back makes a new project in the first empty slot.

Cards A–L: the 12 project slots, A–D on the top row. A filled card is the loaded project, an outlined one has a project, a grey one is empty. L7 starts where you left it: in the looper, or in the menu on the loaded project's card (or the first empty card when none is loaded), with the clock stopped.

The list: export to Ableton, storage mode, the settings, and help: 12 pages on how L7 works. Turn them with the arrows; **A** takes you back. L7 remembers which pages you've seen.

ON A CARD	WHAT IT DOES

A or B	Load that project, as soon as you press; the menu stays open (keep holding for a moment to go to the looper too, starting the clock if it's stopped). While playing, it comes in on the next bar line: until then its card flashes, and the old project already shows as not loaded. A project with no clips (a new one too) comes in at once, the clock starting again from the top. On an empty slot: start a new project and go to the looper when you let go. On the project that's already loaded: go back to the looper when you let go. Holding the button a moment (on any card) goes to the looper without letting go, and also starts the clock if it's stopped. While a project you loaded is still on its way in, you stay in the menu; pressing it again (or holding) takes you to the looper as soon as it starts playing.
SHIFT + A	Cut the project to the project clipboard (hold for a second: the card wipes away as you hold; let go of either key sooner and nothing happens), or paste it into an empty slot. Cutting a project with no clips removes it but leaves the clipboard as it was.

START starts and stops the clock in the menu too. Each project keeps its own scenes and tempo; new projects start with the tempo you last set by hand. Each project keeps its own master volume and balance too; new projects start at 100%. Opening the menu with a project that has no clips at all deletes that project.

Settings

Choose **SETTINGS** in the menu. A small window opens above it. Use UP / DOWN to pick a line and A to turn it on or off. DOWN past the bottom line, or LEFT / RIGHT, closes it. L7 remembers your settings. All are on at first.

- **Monitor audio in**: you hear the input through L7. Off, you don't, but it still records.
- **Play newly recorded**: a new loop plays when you finish it. Off, it comes in muted.
- **Delete unused samples**: press A to empty the clipboards, then delete from the card every recording that no project, scene or track uses, and every silent recording (taken off the scenes and tracks that had it) (the clock stops first). A bar fills the line as it goes, then it says how many samples were deleted. Once that message goes, the item becomes **Defrag samples**: press A and every recording stored in pieces on the card is copied into one unbroken stretch of free space and put back in its place (it says how many, and how many found no room; "ALL IN ONE PIECE" if there was nothing to do); this can help with dropouts on a well-used card. Then it's back to Delete unused samples.
- **Select next free track**: when you stop a recording (or after a resampling), the next empty track is selected (none empty: the next muted one). Off, you stay on the track you recorded.
- **Empty project click**: a project with no loops yet clicks with the clock, even with the metronome off.

When you leave the menu by pressing A on a card, coming back puts you on the loaded project's card.

Exporting to Ableton Live

Load the project, open the menu and choose **EXPORT ABLETON SET**. You can do it while playing, straight after recording too. The clock stops and L7 writes a Live 11 project to the card, in `/L7/EXPORTS/L7 A01 Project/` (the project's letter, then the next free number, 01–99), the label counting as it goes (**EXPORTING 3/7**). The label then reads **SAVED AS L7 A01**, and the button stays off until you close the menu. If the clock was running, it starts again from 1.1.1.

- 7 audio tracks, coloured like the project, each with a highpass and a lowpass Auto Filter and a Utility.

- One scene per L7 scene that has clips, each with its L7 scene's tempo (Live switches to it when the scene launches), plus an empty one at the end. The current scene's clips are armed to play when the set opens.
- Each clip keeps its loop range, start, mute, volume (as clip gain, with the filter's level make-up), balance and filter settings (as clip envelopes).
- The samples are copied into the project, so you can move it anywhere. A clip whose range runs across the end of its loop gets a second copy of its sample with the loop written twice (`A001x2.wav`), so Live can play the range without wrapping.

Storage mode: the card on your computer

In the menu, choose **STORAGE MODE**, then connect the M8 to your computer with a USB cable. The SD card mounts as a normal drive. The clock stops, and the screen shows the connection state.

When you're done, eject the drive on the computer, or press any key on the M8. L7 reads the card again and you're back in the menu.

> Always eject or press a key before unplugging. Files you add, rename or delete on the computer are picked up when you leave storage mode.

What's on the card

FOLDER	CONTENTS
`/L7/SAMPLES`	Every recording, as a WAV file named `A001.wav`: the project letter and a number counting up within the project (001–999). Older recordings named like `A30012.wav` (with the track number) keep their names.
`/L7/PROJECTS`	The projects, `A.L7P` to `L.L7P`. An empty slot has no file.
`/L7/EXPORTS`	The Ableton projects you exported.
`/L7/DATA`	Everything else L7 needs, including about 180 MB of space kept ready for new recordings.

Recordings stay on the card even when no project uses them any more.

Good to know

- L7 saves as you go: tracks, scenes, tempo and the loaded project survive switching off.
- When an action can't run right now (for example paste with an empty clipboard), the action button turns grey and **A** does nothing.
- The headphone output mutes the speaker automatically.
- With the clock stopped you can swap the SD card: take it out and the screen asks for one (**INSERT AN SD CARD**); put one in and L7 reads its projects and recordings (a blank card gets a fresh `/L7` folder). Switched on without a card, it waits for one the same way.
- MIDI clock and start/stop go out on the M8's MIDI out.

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